CST8507: NATURAL LANGUAGE PROCESSING

WEEK#11

LLM & RAG

DEVELOPED BY

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Lesson Agenda

- Introduction to LLM
- LLMs Components
- How ChatGPT trained
- Limitations of LLMs
- How Retrieval Augmented Generation **RAG** works
- LangChian

What is LLMs

- Trained on **massive amounts of text data**.
- Transformer architecture
- Understanding and generating human-like texts.
- Capable of performing a wide range of NLP tasks with high proficiency, including text completion, summarization, ….

A Timeline Of Existing Large Language Models

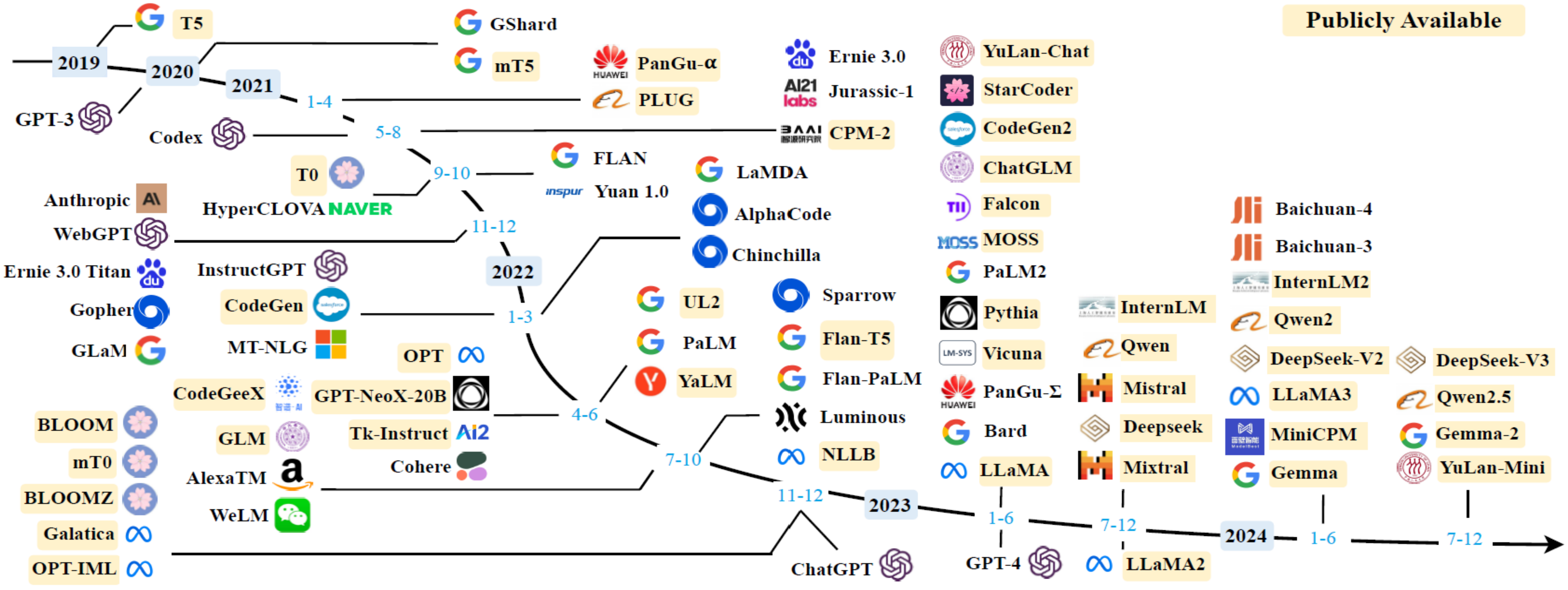

Image sources:https://www.marktechpost.com/2024/11/09/a-deep-dive-into-small-language-models-efficient-alternatives-to-large-language-models-for-real-time-processing-and-specialized-tasks/, Wayne Xin Zhao elt

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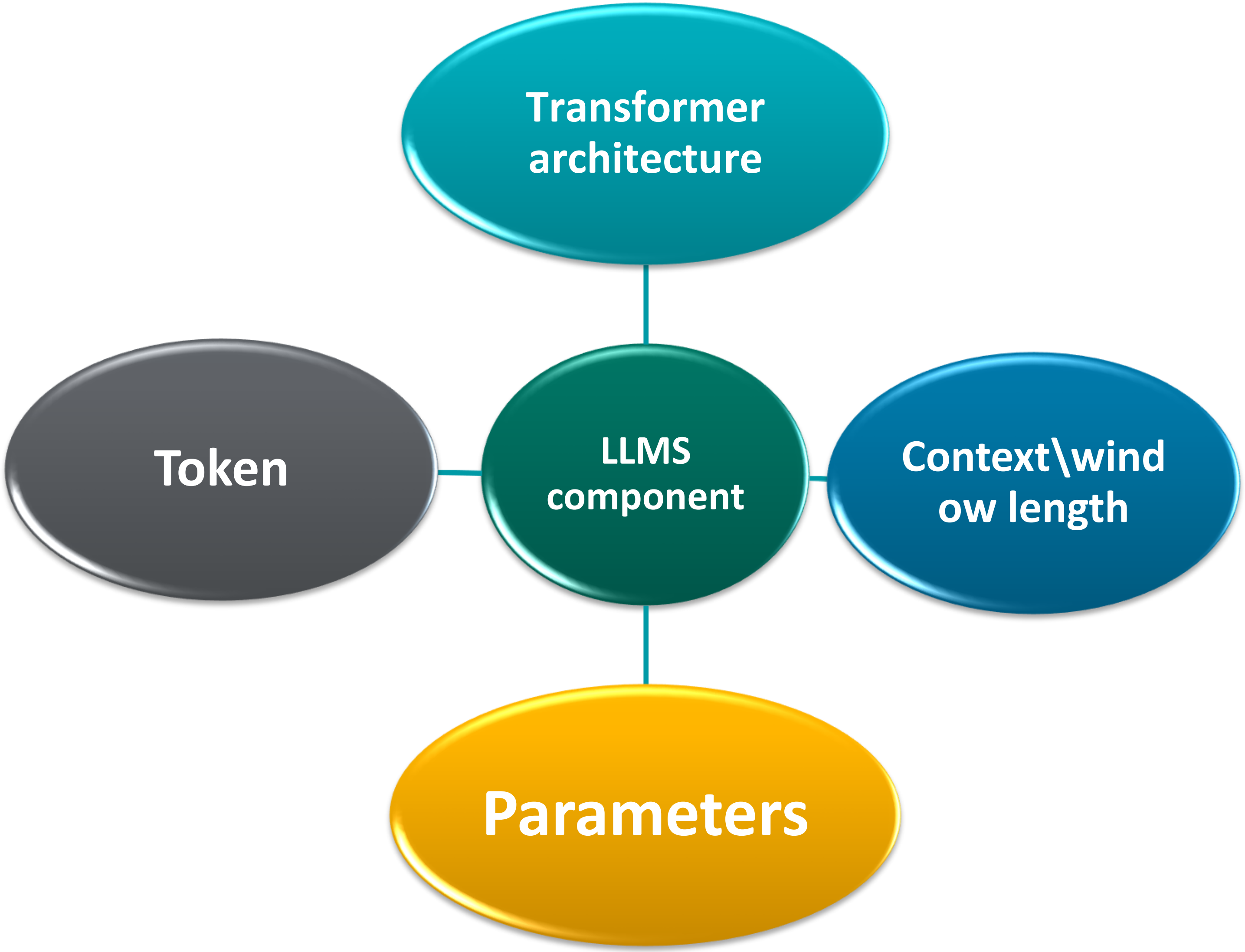
LLMs Components



Comparison of Popular Large Language Models

Model	Parameters	Size on Disk	Memory Usage (Inference)	Learning Data Size
BERT (Large)	340M	~1.3 GB (FP32)	~1.5–2 GB (FP16)	3.3B words (~16 GB)
GPT-4o	~200B	~350 GB (FP32)	~400 GB (FP16, single GPU)	570 GB (~300B tokens)
LLaMA (13B)	13B	~26 GB (FP32)	~26 GB (FP16)	~1T tokens
LLaMA (70B)	70B	~140 GB (FP32)	~140 GB (FP16)	~1T tokens
BLOOM (176B)	176B	~352 GB (FP32)	~352 GB (FP16)	1.6T tokens
Mistral 7B	7B	~14 GB (FP32)	~14 GB (FP16)	~1T tokens
Mixtral 8x7B	56B	~112 GB (FP32)	~112 GB (FP16)	Unknown (large corpus)
Grok (xAI)	Unknown (est. ~70B)	Est. ~140 GB (FP32)	Est. ~140 GB (FP16)	Unknown (large)
PaLM (540B)	540B	~1 TB (FP32)	~1 TB (FP16)	780B tokens

LLM Tokenizer

https://tiktokenizer.vercel.app/

Context Length

The **maximum number of tokens** an LLM can process at once.

Includes:

- prompt
- conversation history
- generated answer

Context Length :Comparison of Various LLMs

Types of large language models

- A general-purpose language model(GPT series)
- instruction tuned language models(AllenNLP's)
- dialogue-tuned language models(Microsoft's DialoGPT)
- Domain specific language models (BioBERT),Bidirectional Encoder Representations from Transformers for Biomedical Text Mining

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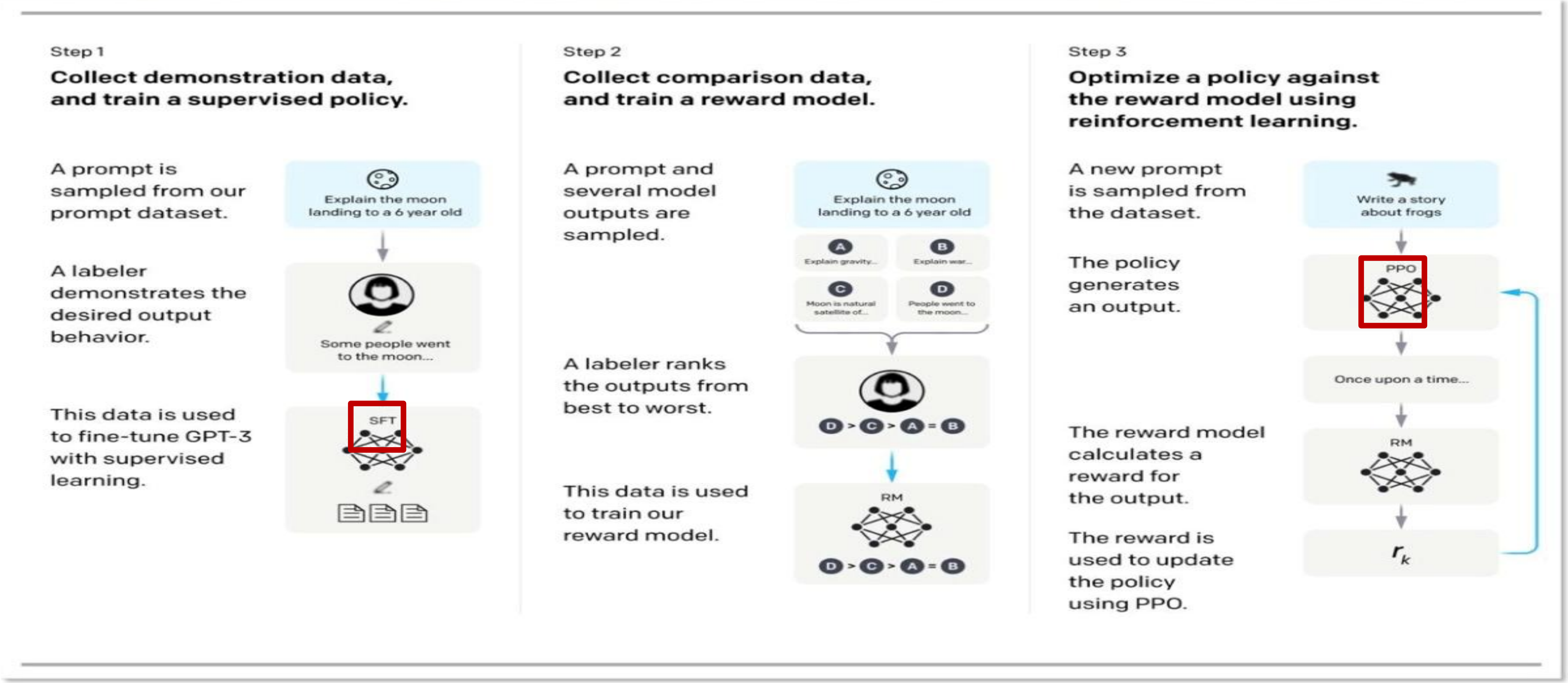
How ChatGPT is Trained



The Modern Upgrade: (Claude and GPT4o)

Constitutional AI (CAI)

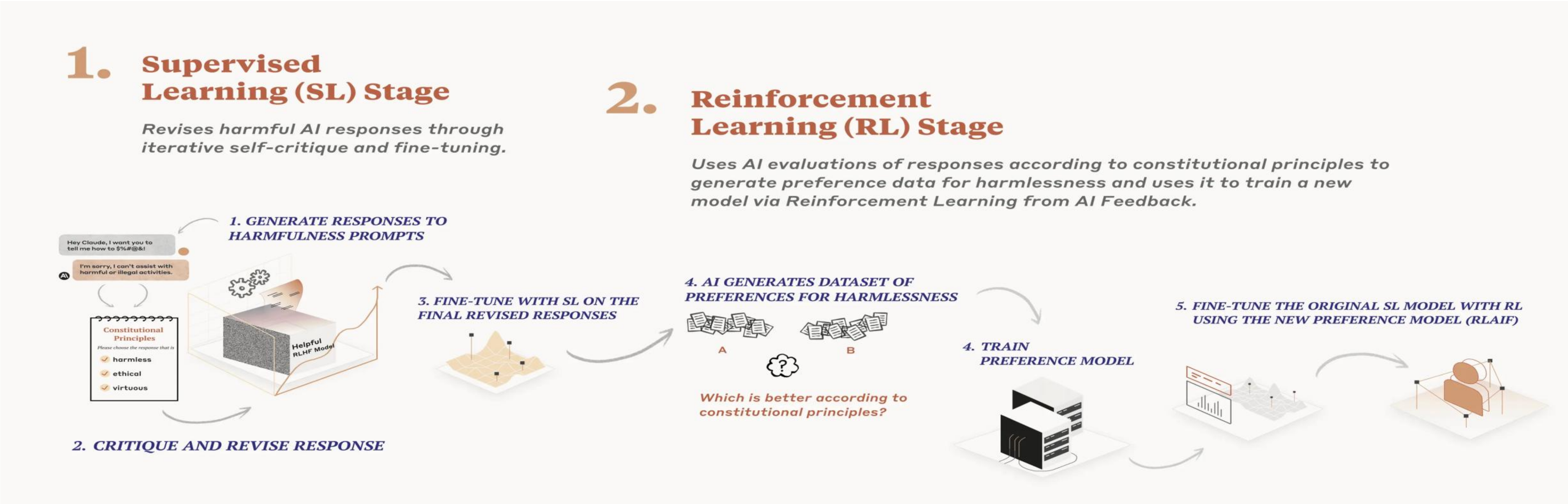


Image source: https://www.anthropic.com/news/claudes-constitution

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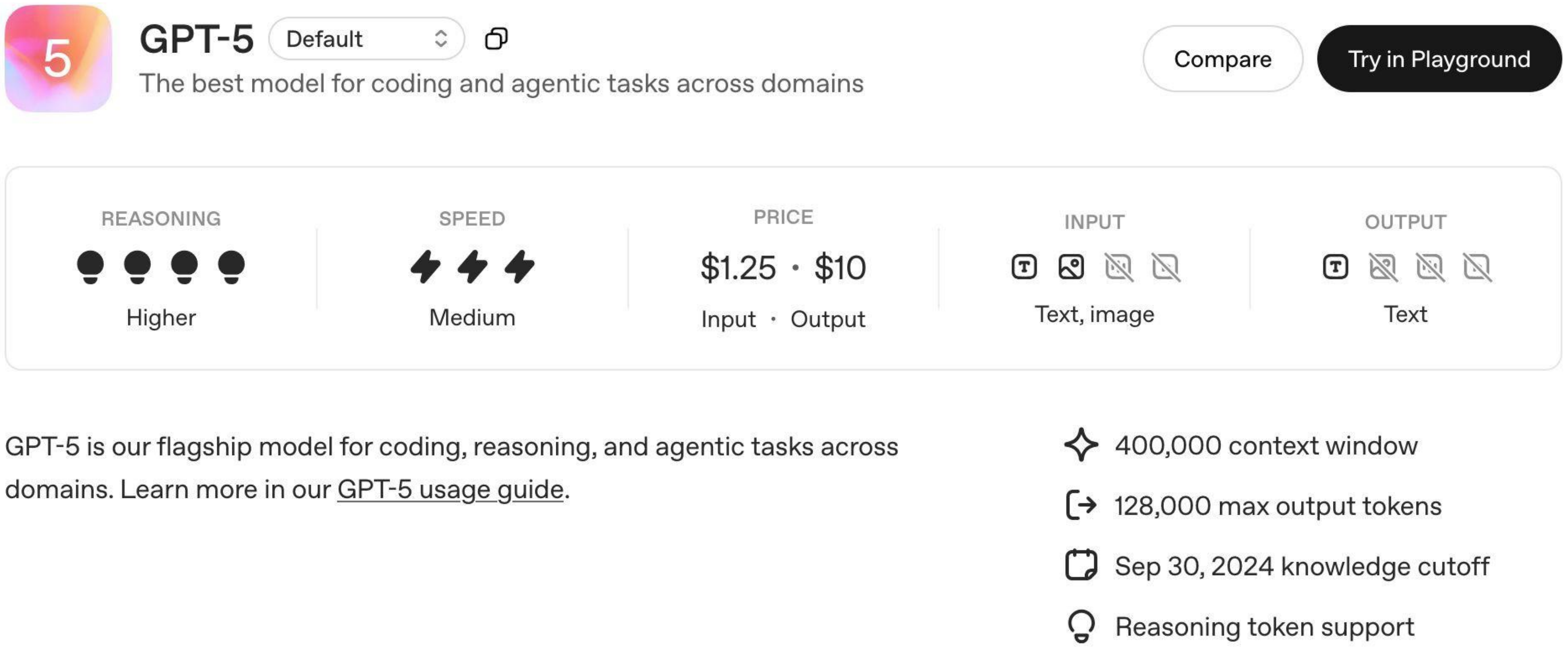
LLM Limitations

- **Knowledge** of LLM **constrained** to pretraining data



Screenshot of OpenAI's [GPT-5 model card](), taken on November 9th, 2025.

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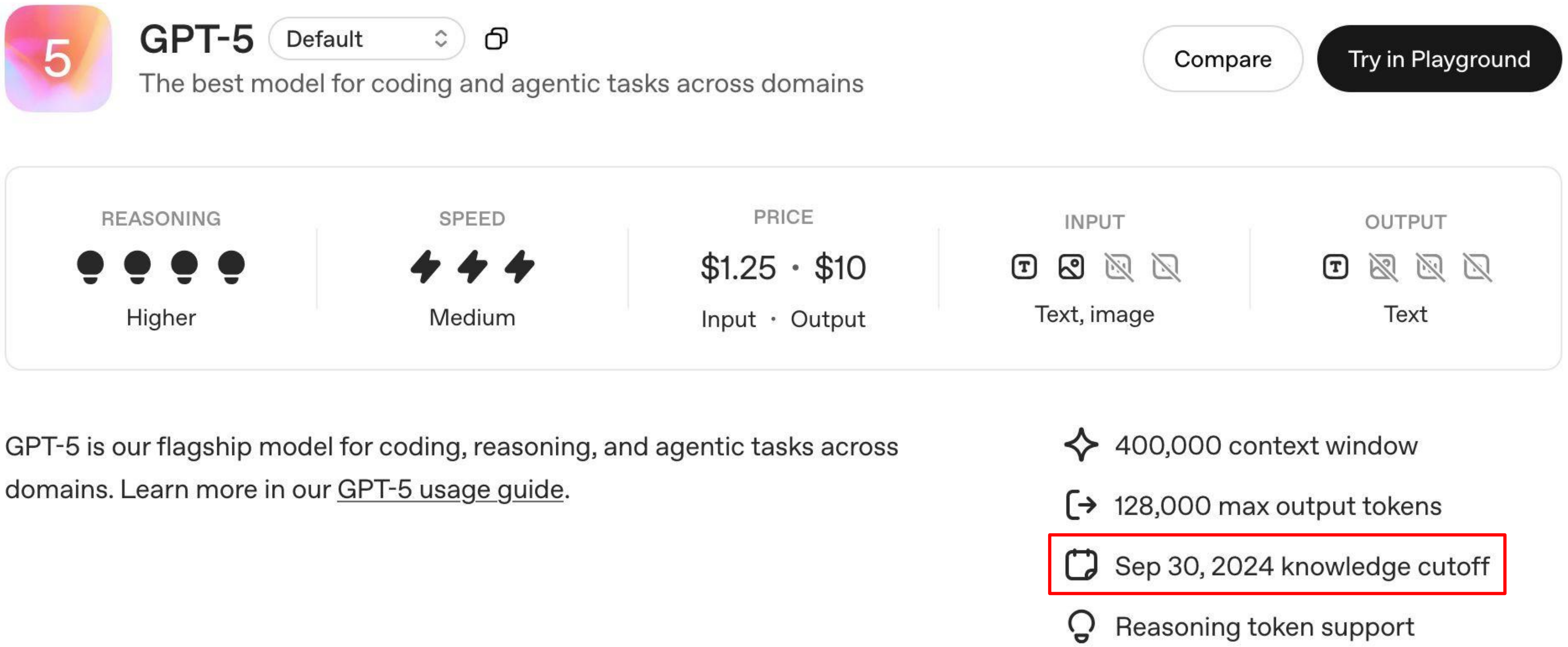
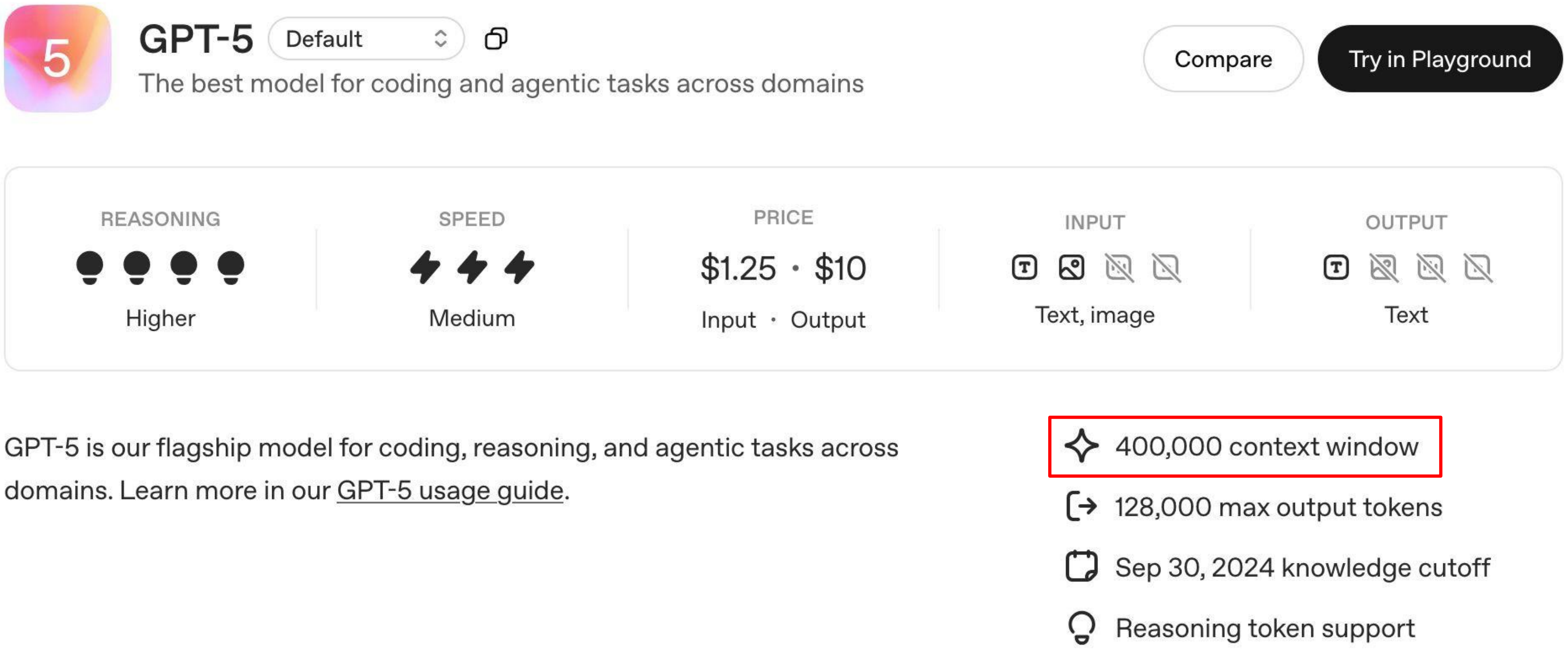


Screenshot of OpenAI's [GPT-5 model card](), taken on November 9th, 2025.

LLM Limitations

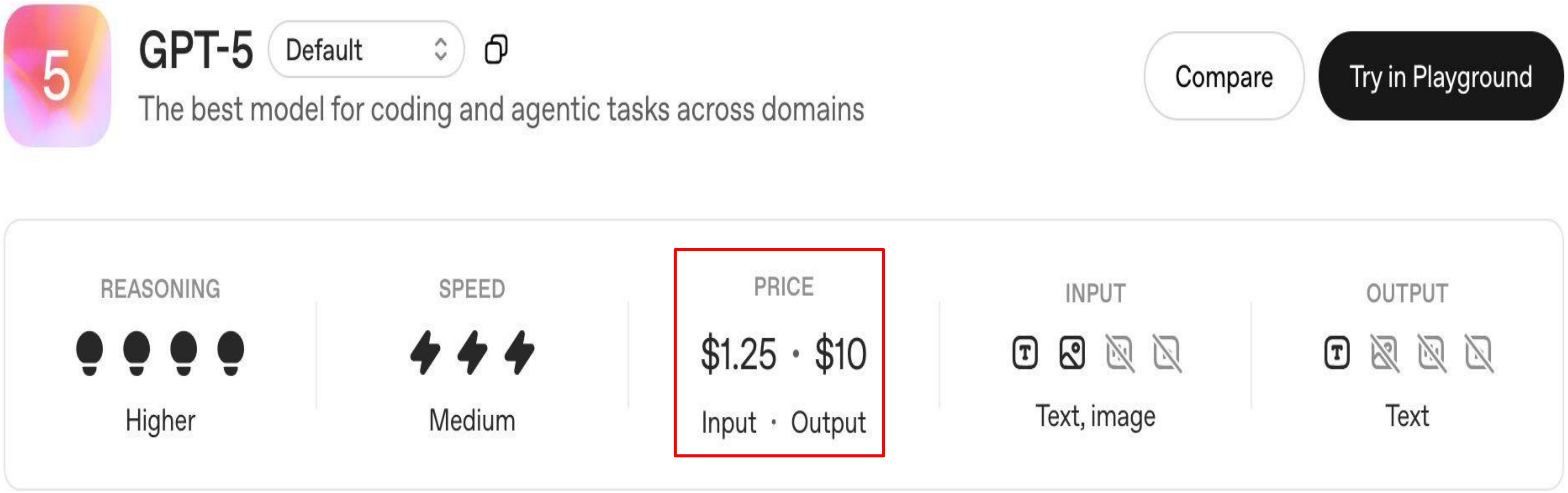
- **Limited** context size



Screenshot of OpenAI's [GPT-5 model card](), taken on November 9th, 2025.

LLM Limitations

- **Pricing** is **per** input/output **token**



Other LLM Challenges

- Hallucination
- Lack of specialized information
- Lack of transparency
- Privacy & Security Risks
- Energy Consumption & Environmental Impact
- High Computational & Memory Costs

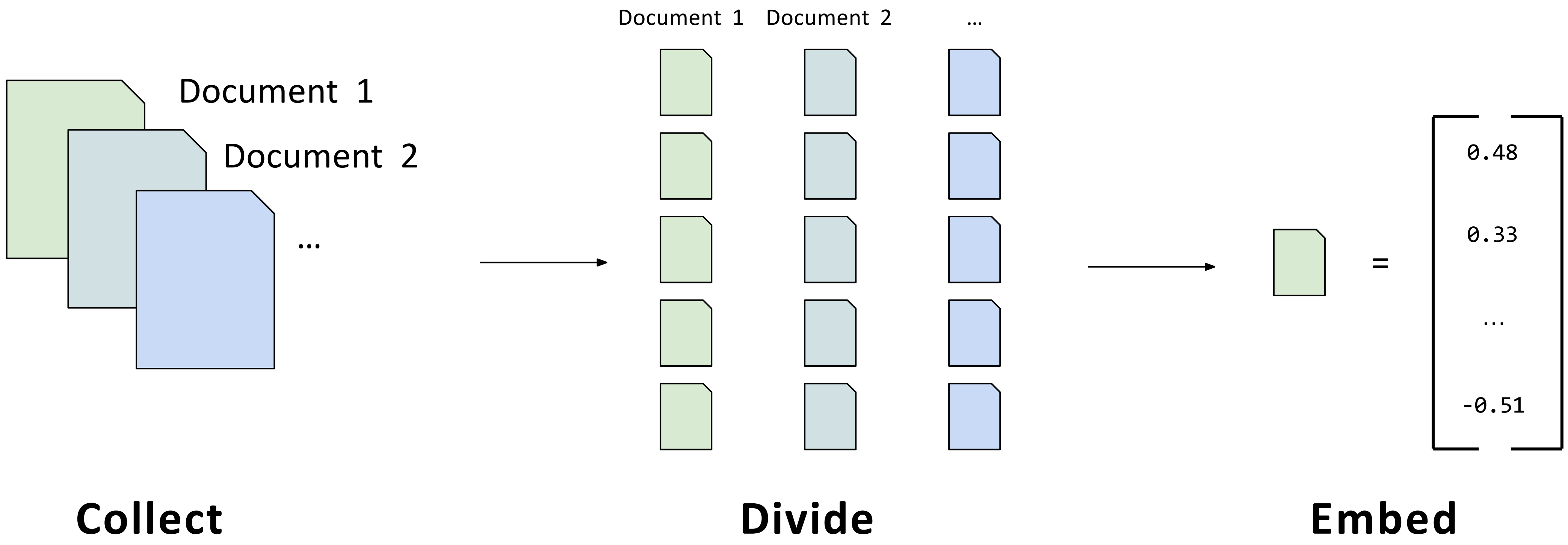
Galactica

RAG: Motivations

RAG = Retrieval-Augmented Generation

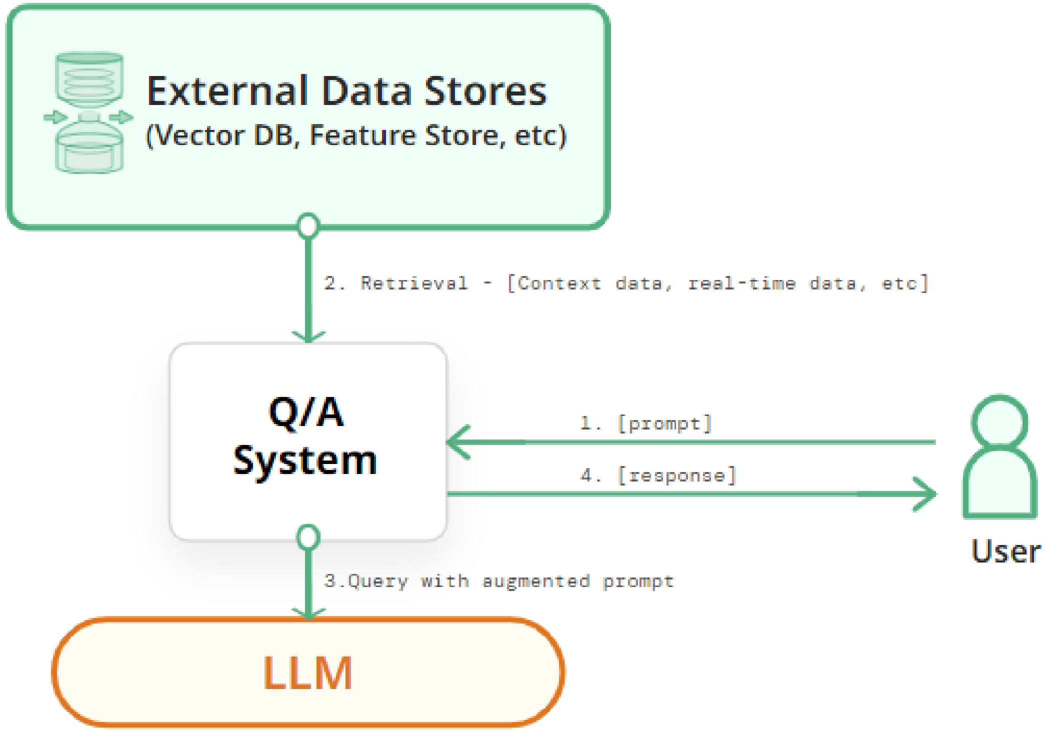
- Domain-specific accurate answering
- Frequent updates of data
- Traceability and explainability of generated content
- Controllable Cost
- Privacy protection of data

Prerequisite: Create knowledge base



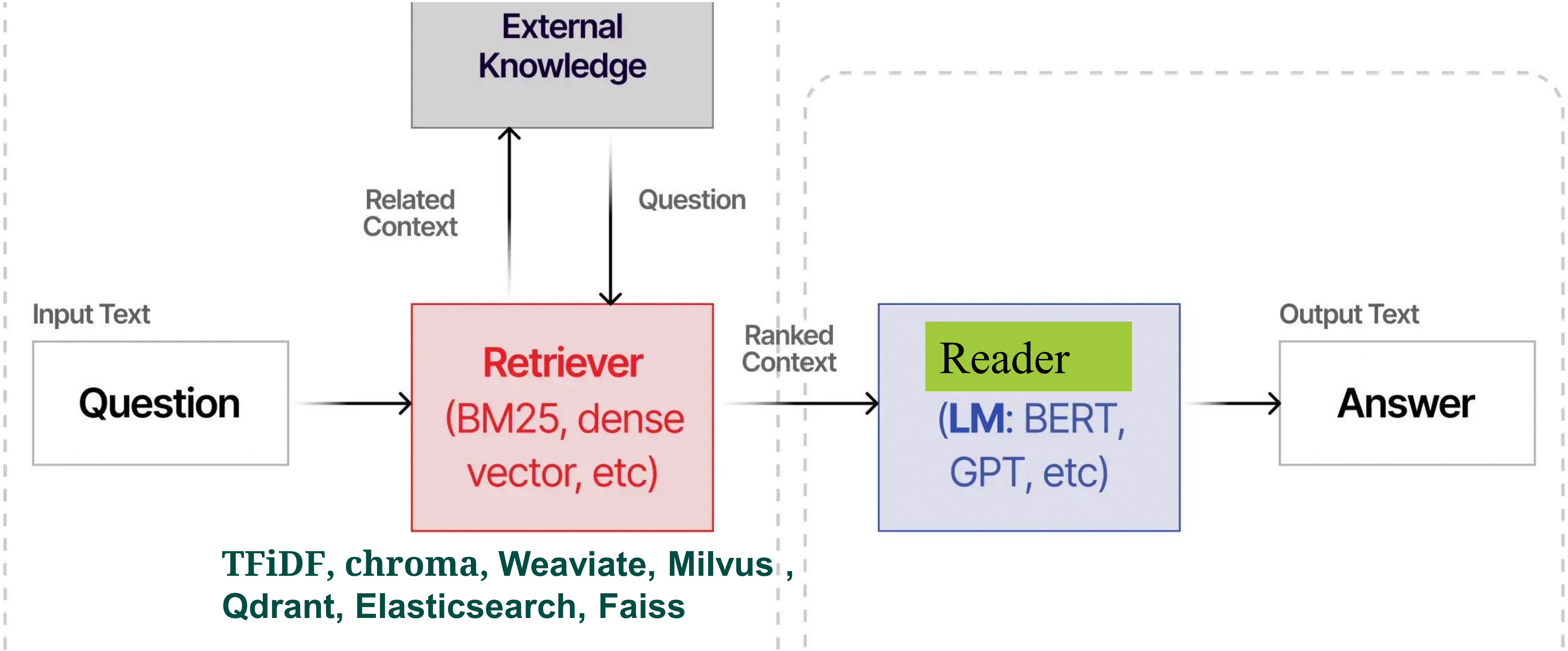
RAG systems

- Retrieval augmented generation **RAG**: Augmented LLM with specialized and mutable knowledge base



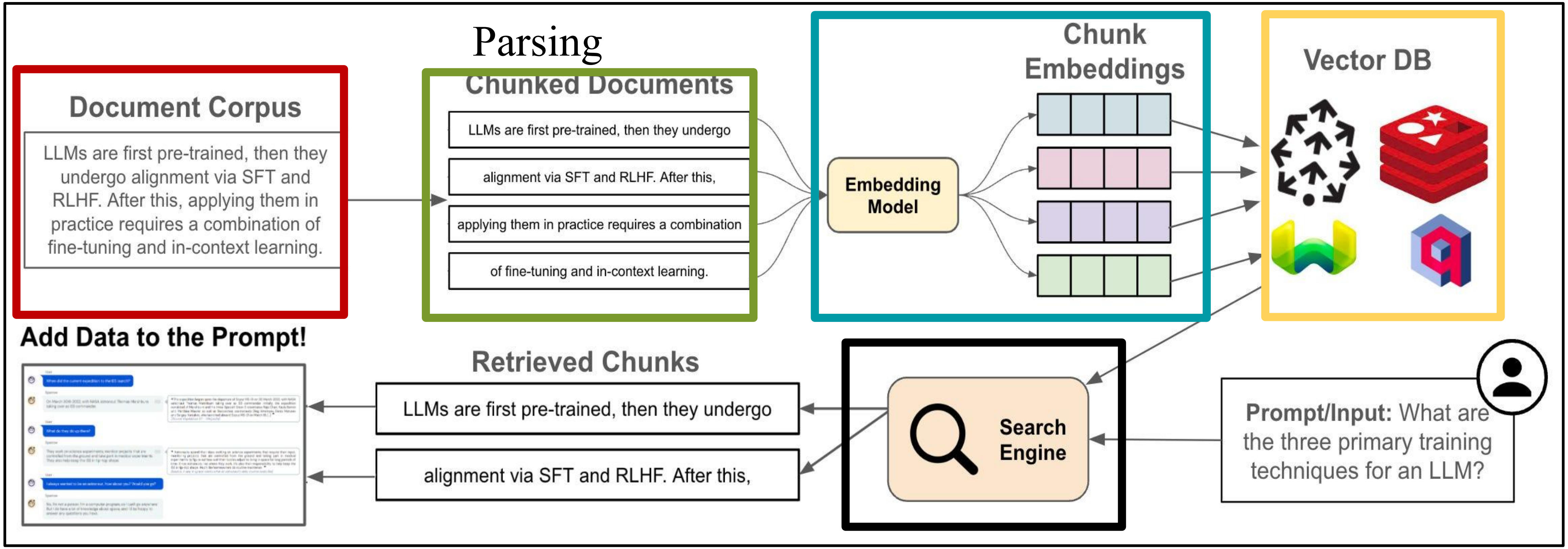
[“Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”](#) (2020)

How RAG works



How RAG works…

Indexing

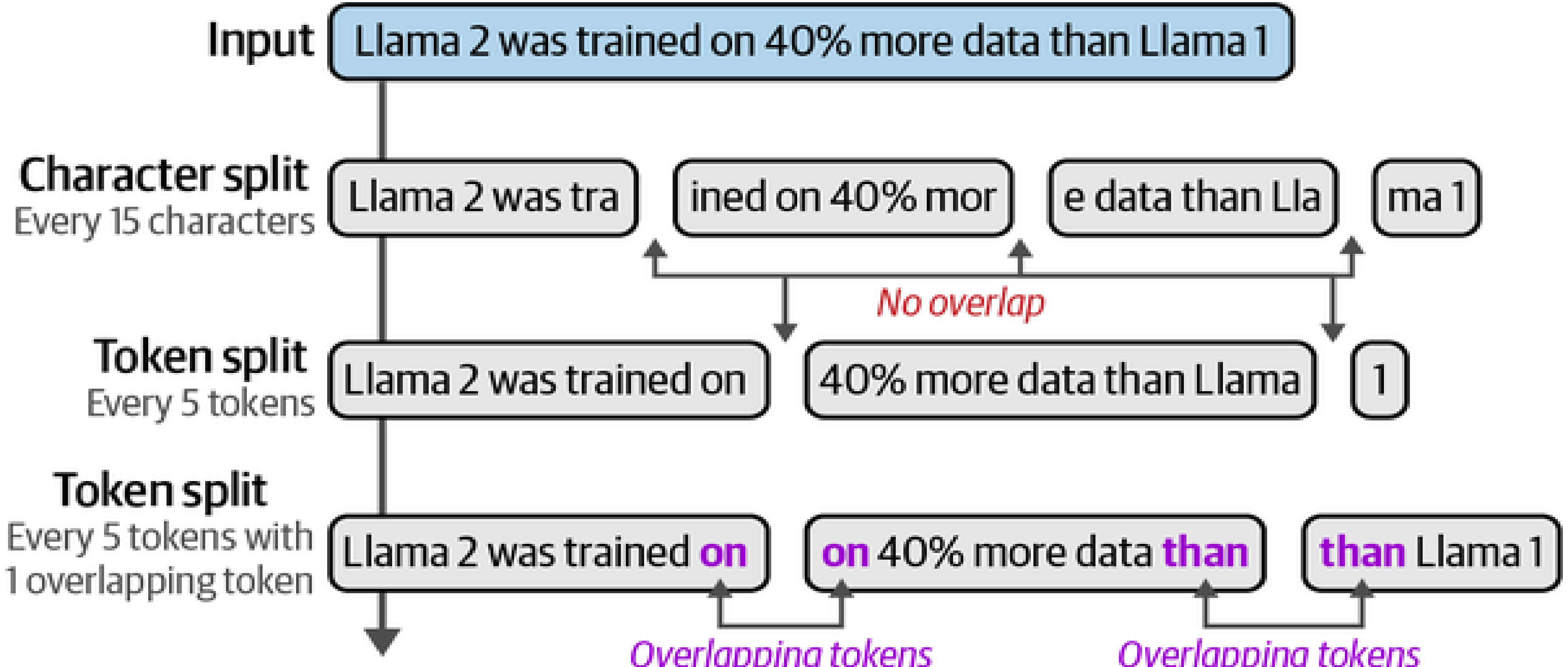


Semantic search

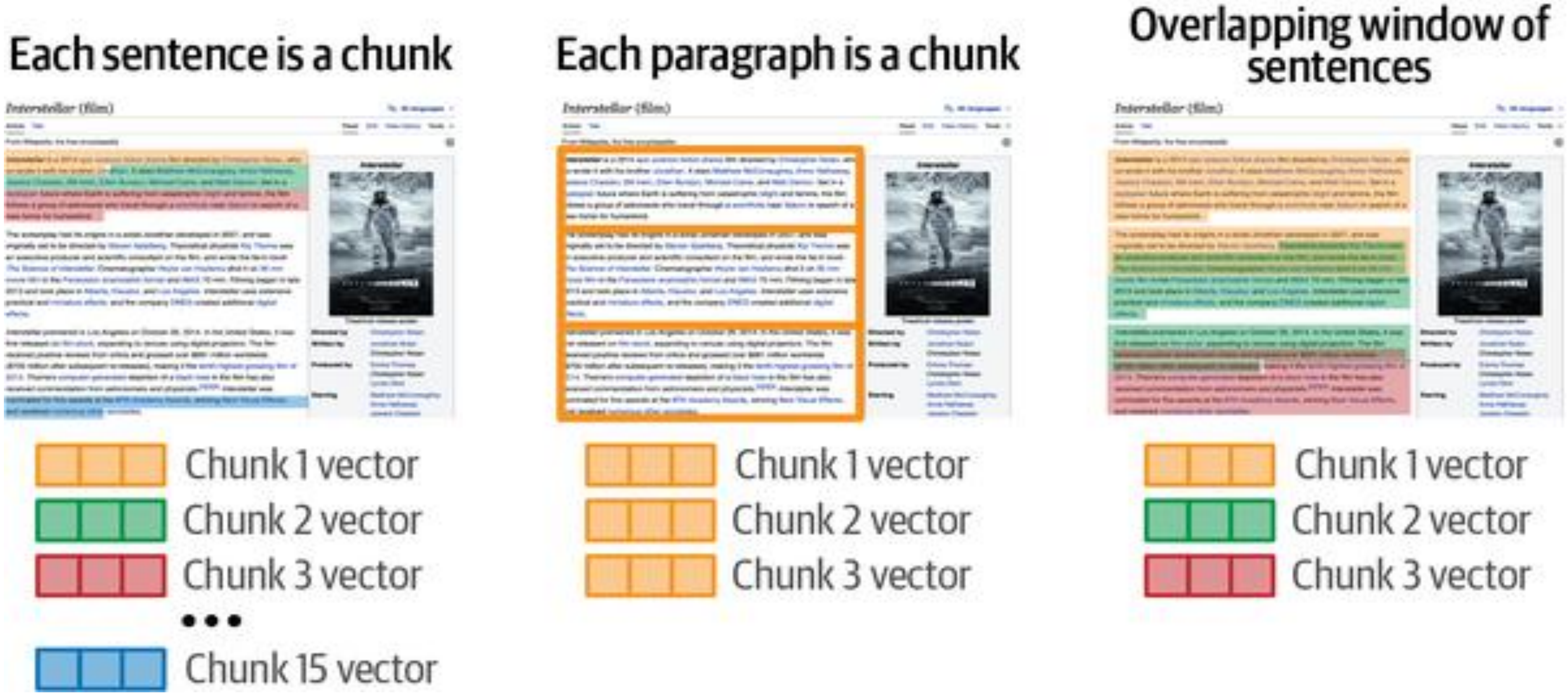
For More information about RAG with LLM

https://cameronrwolfe.substack.com/p/a-practitioners-guide-to-retrieval

Chunking Methods(Fixed Size Chunking)



Chunking Methods(split on natural boundaries)



Prompt_Templet

```
Prompt_Templet="answer the question based only the
following context:{context}
---
Answer the question based on the above context :{question}"
```

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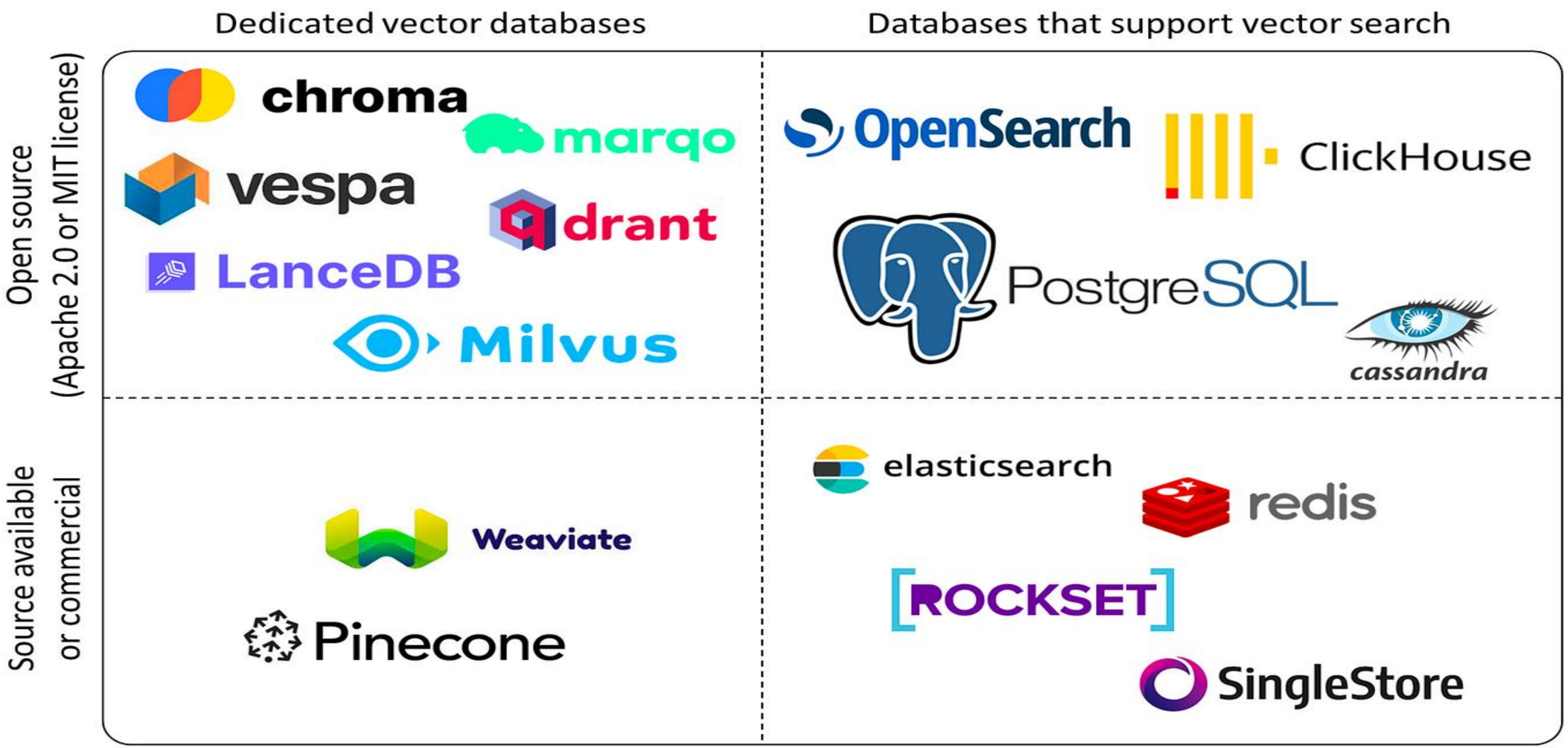
Vector database



https://blog.det.life/why-you-shouldnt-invest-in-vector-databases-c0cd3f59d23c

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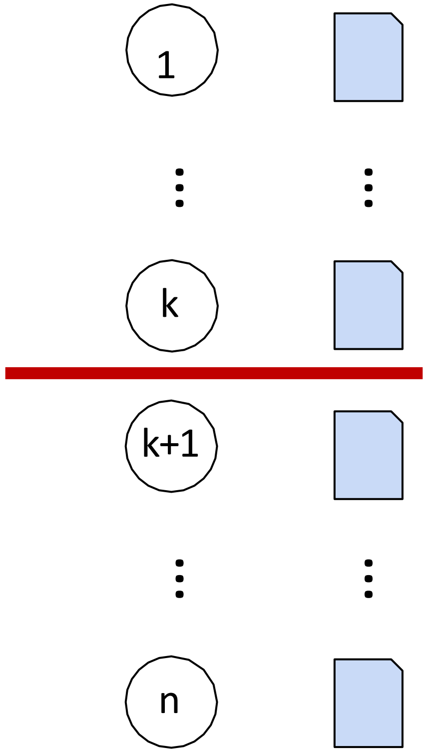
Quantify performance of retrieval

Ranking



- **N**ormalized **D**iscounted **C**umulative **G**ain at **k** (**NDCG@k**)

$$\boxed{\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{\mathrm{rel}_i}{\log_2(i+1)}}$$

rel_i =relevance score at position i

•$\log_2(i+1)$ =penalty for lower ranks

with $\mathrm{rel}_i \in \{0, 1\}$

$$\mathrm{NDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k}$$

$\mathrm{DCG@}k$ if ranking was perfect

NDCG@k: Example

$$\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{\mathrm{rel}_i}{\log_2(i+1)}$$

Rank	Chunk	Relevance (rel_i)
1	Chunk text 1	1
2	Chunk text 2	3
3	Chunk text 3	2
4	Chunk text 4	0
5	Chunk text 5	1

DCG

1/log2(2)=1
3/log2(3)=1.89
2/log2(4)=1
0/log2(5)=0
1/log2(6)=0.39
Total DCG ≈ 4.28

The ideal Sorted relevance: [3,2,1,1,0]

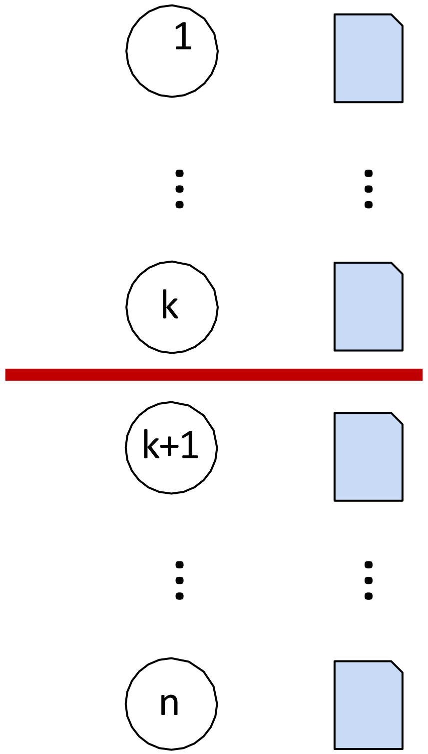
IDCG

3/1=3
2/1.585=1.26
1/2=0.5
1/2.32=0.43
Total IDCG ≈ 5.19

nDCG = DCG / IDCG = 4.28 / 5.19 ≈ 0.82

Quantify performance of retrieval…

Ranking



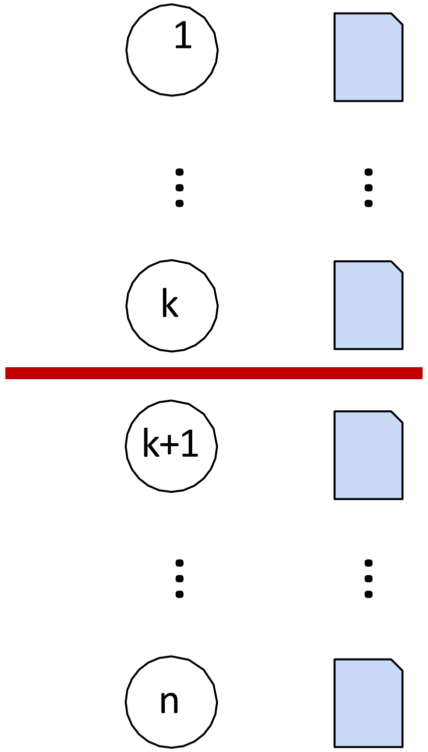
- Reciprocal Rank at **k** (**RR@k**)

$$\mathrm{RR} = \frac{1}{\mathrm{rank}}$$

rank of the first relevant chunk

Quantify performance of retrieval…

Ranking

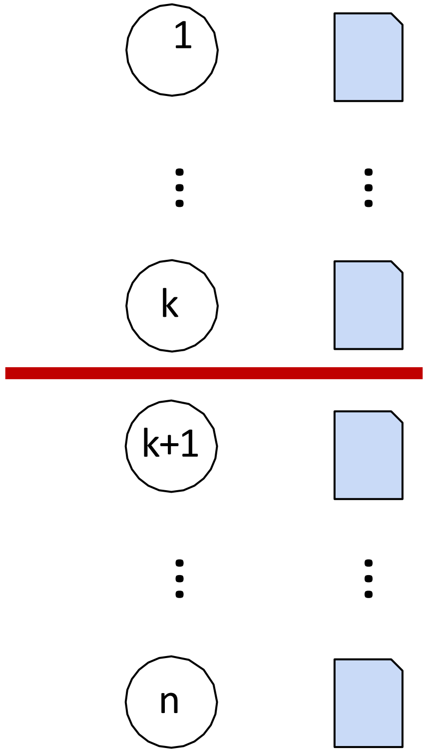


- **Recall at k**

$$\text{Recall@}k = \frac{|\text{relevant in top } k|}{|\text{relevant}|}$$

Quantify performance of retrieval…

Ranking



- **Precision at k**

$$\text{Precision@}k = \frac{|\text{relevant in top } k|}{k}$$

Settings to keep in mind

- One important setting is controlling how deterministic the model is when generating completion for prompts
  - **Temperature** and **top_p** are two important parameters to keep in mind
  - Generally, keep these low if you are looking for exact answers
  - keep them high if you are looking for more diverse responses

RAG Frameworks

- There are many tools, libraries, and platforms with different capabilities and functionalities
- Capabilities include:
  - Developing and experimenting with prompts
  - Evaluating prompts
  - Versioning and deploying prompts

haystack

Dyno
Prompt Engineering IDE

LangChain

LlamaIndex

DUST

PROMPTABLE

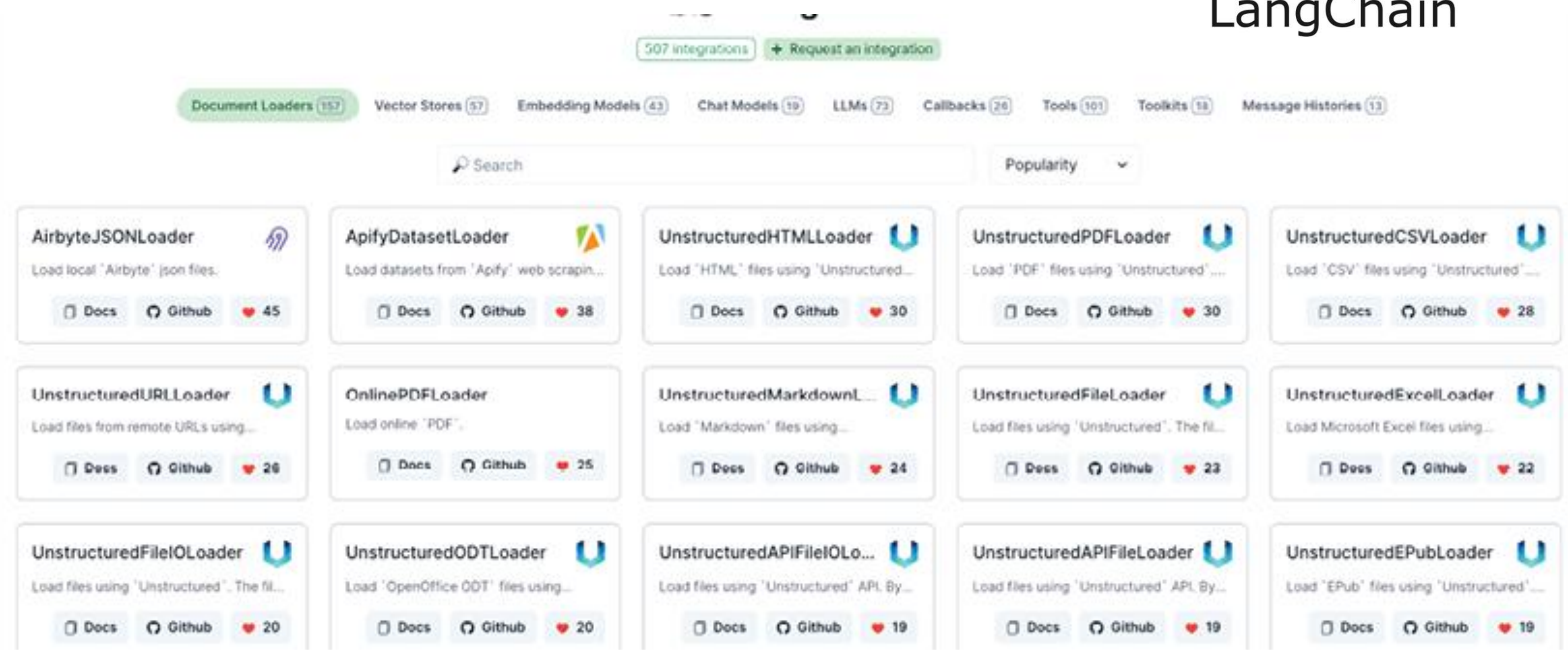
More tools here: https://github.com/dair-ai/Prompt-Engineering-Guide#tools--libraries

Lang Chain framework

- **Modular architecture** for flexible and adaptable LLM integrations.
- **Chaining together** multiple services beyond just LLMs.
- Goal-driven agent interactions instead of isolated calls.
- **Memory and persistence** for statefulness across executions.
- **Open-source access** and community support.

https://python.langchain.com/docs/get_started/introduction

Lang Chain…


507 integrations
+ Request an integration
Document Loaders 157
Vector Stores 57
Embedding Models 43
Chat Models 19
LLMs 73
Callbacks 26
Tools 101
Toolkits 18
Message Histories 13
Search
Popularity
AirbyteJSONLoader
Load local `Airbyte` json files.
Docs Github 45
ApifyDatasetLoader
Load datasets from `Apify` web scrapin...
Docs Github 38
UnstructuredHTMLLoader
Load `HTML` files using `Unstructured...
Docs Github 30
UnstructuredPDFLoader
Load `PDF` files using `Unstructured`....
Docs Github 30
UnstructuredCSVLoader
Load `CSV` files using `Unstructured`....
Docs Github 28
UnstructuredURLLoader
Load files from remote URLs using...
Docs Github 26
OnlinePDFLoader
Load online `PDF`.
Docs Github 25
UnstructuredMarkdownL...
Load `Markdown` files using...
Docs Github 24
UnstructuredFileLoader
Load files using `Unstructured`. The fil...
Docs Github 23
UnstructuredExcelLoader
Load Microsoft Excel files using...
Docs Github 22
UnstructuredFileIOLoader
Load files using `Unstructured`. The fil...
Docs Github 20
UnstructuredODTLoader
Load `OpenOffice ODT` files using...
Docs Github 20
UnstructuredAPIFileIOLo...
Load files using `Unstructured` API. By...
Docs Github 19
UnstructuredAPIFileLoader
Load files using `Unstructured` API. By...
Docs Github 19
UnstructuredEPubLoader
Load `EPub` files using `Unstructured`....
Docs Github 19

OPEN Source MLL

- Core general-purpose models

Meta Llama 3

MISTRAL AI_

Qwen

Gemma 3

Deepseek R1

Microsoft Phi-4

OPEN Source MLL

💻 8 GB RAM (Laptop / Low-End PC)

Model	Size on Disk	Ollama Command
Llama 3.3 8B	4.9 GB	`ollama pull llama3.3:8b`
Mistral 7B	4.1 GB	`ollama pull mistral:7b`
DeepSeek R1 7B	~5 GB	`ollama pull deepseek-r1:7b`
Phi-4	~4 GB	`ollama pull phi4`

https://wp.astera.com/type/blog/open-source-vs-closed-source-llms/

OPEN Source MLL…

🖥️ **16 GB RAM (Mid-Range Workstation)**

Model	Size on Disk	Ollama Command	Q&A Strength
DeepSeek R1 14B	~9 GB	ollama pull deepseek-r1:14b	Chain-of-thought Q&A
Qwen 2.5 14B	~9 GB	ollama pull qwen2.5:14b	Multilingual Q&A
Phi-4 14B	~8 GB	ollama pull phi4:14b	Analytical reasoning

Choosing a Platform to Run LLMs

Tools to Run and
Host Open-source LLMs

Ollama

vLLM

Hugging Face

LM Studio

Comparison between Ollama, vLLM, Hugging Face

Factor	Ollama	vLLM	Hugging Face
Ease of Use	Very user-friendly	Requires more technical setup	User-friendly with extensive docs
Performance	Optimized for smaller models	Highly optimized for large-scale models	Good performance with scaling options
Scalability	Limited	High	Scalable with managed services
Integration & Flexibility	Limited flexibility	High flexibility for advanced users	High flexibility, integration into multiple tools
Model Support	Popular open-source models	Popular open-source models	Extensive (Transformers, etc.)
Deployment	Local	Cloud/On-premise	Cloud/On-premise
Use Case	Ideal for small to medium-scale	Best for high-throughput, enterprise-level use	Great for research, fine-tuning, and both small and large-scale use

LLM Models open source

https://ollama.com/library

https://lmstudio.ai/models

Leaderboard: Community-driven Evaluation for Best LLM

Model 627 / 627	Overall	Expert	Hard Prompts	Coding	Math	Creative Writing	Instruction Following	Longer Query
claude-opus-4-6-thi...	1	2	1	1	2	1	1	1
claude-opus-4-6	2	1	2	2	3	5	2	2
gemini-3.1-pro-prev...	3	4	3	3	4	4	3	3
grok-4.20-beta1	4	22	5	6	17	2	10	13
gemini-3-pro	5	10	7	11	5	3	8	6
gpt-5.4-high	6	3	4	4	1	10	4	7
grok-4.20-beta-0309...	7	12	6	7	16	7	16	18
gpt-5.2-chat-latest...	8	16	8	8	12	15	11	15
gemini-3-flash	9	14	12	19	7	9	15	14
claude-opus-4-5-202...	10	7	9	5	8	6	5	4
grok-4.1-thinking	11	25	18	25	28	22	33	30

Source: https://huggingface.co/spaces/lmarena-ai/chatbot-arena-leaderboard

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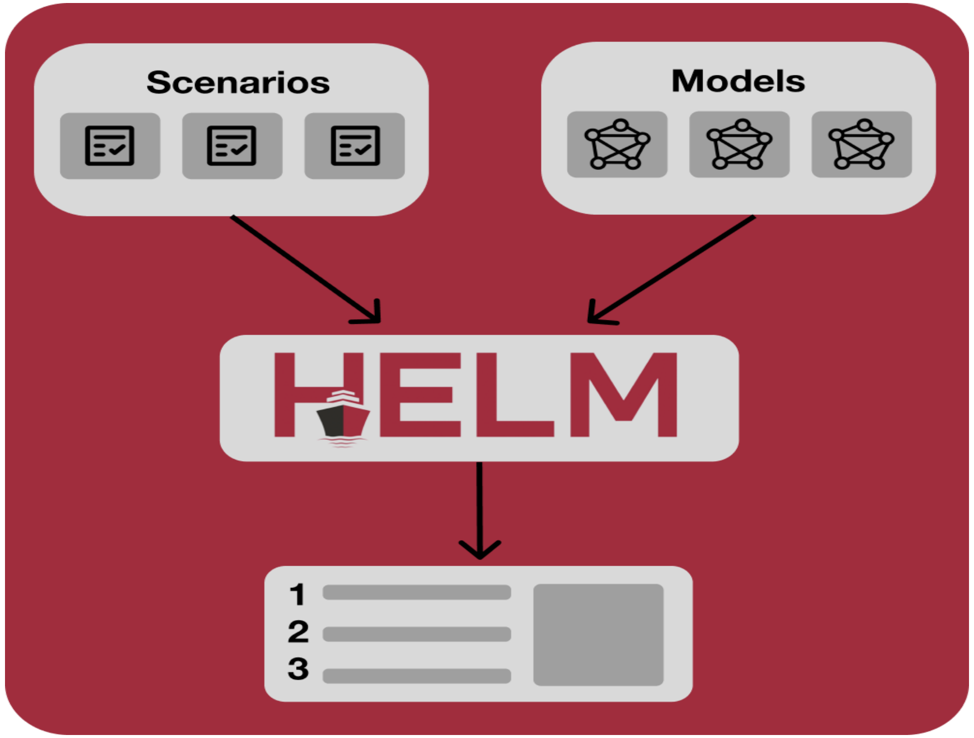
COMPARING DIFFERENT LLM MODELS

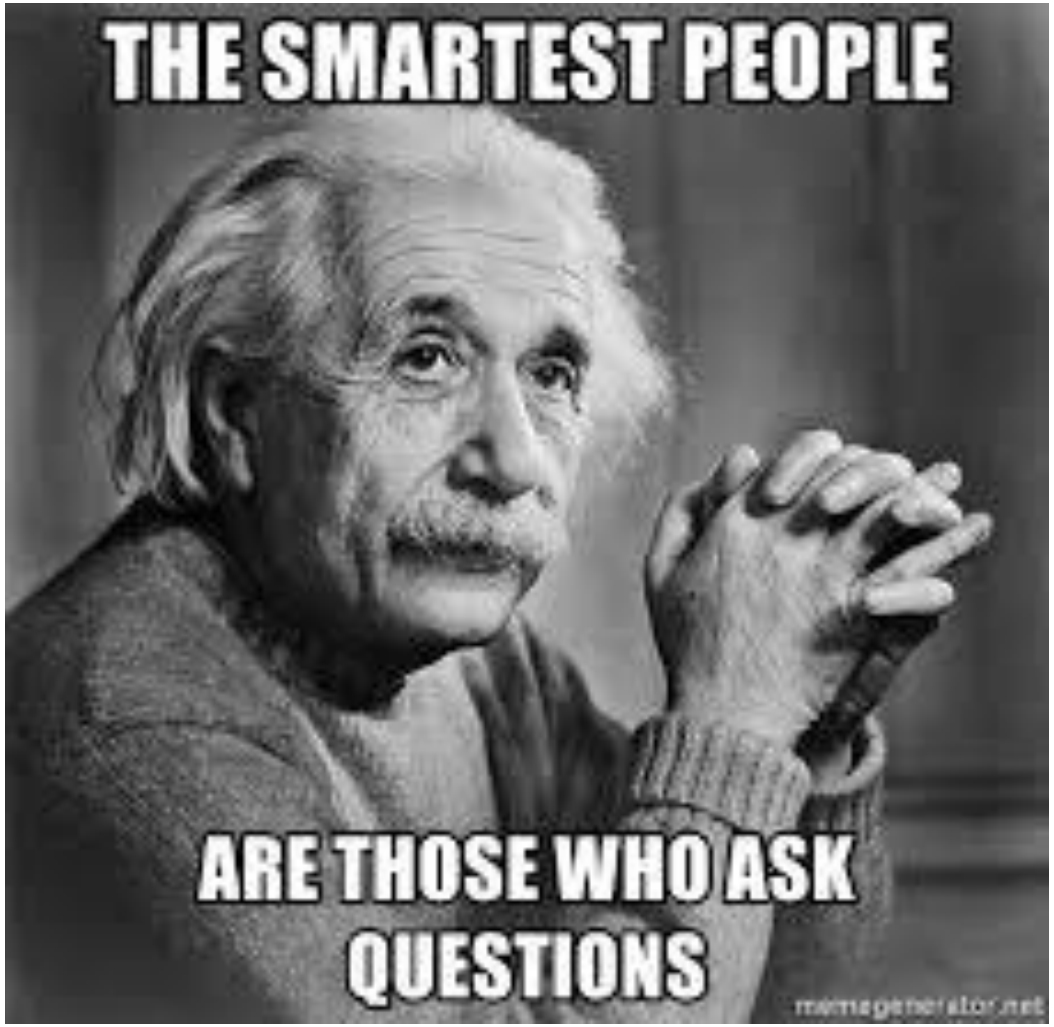


Holistic **E**valuation of **L**anguage **M**odels

HELM Evaluation Dimensions

- Accuracy
- Calibration
- Robustness
- Fairness
- Bias
- Efficiency

Q&A


THE SMARTEST PEOPLE
ARE THOSE WHO ASK QUESTIONS
memegenerator.net